

RESEARCH ARTICLE

Heteroatom tuning in agarose derived carbon aerogel for enhanced potassium ion multiple energy storage

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Abstract

The incorporation of heteroatoms into carbon aerogels (CAs) can lead to structural distortions and changes in active sites due to their smaller size and electronegativity compared to pure carbon. However, the evolution of the electronic structure from single-atom doping to heteroatom codoping in CAs has not yet been thoroughly investigated, and the impact of codoping on potassium ion (K^+) storage and diffusion pathways as electrode material remains unclear. In this study, experimental and theoretical simulations were conducted to demonstrate that heteroatom codoping, composed of multiple heteroatoms (O/N/B) with different properties, has the potential to improve the electrical properties and stability of CAs compared to single-atom doping. Electronic states near the Fermi level have revealed that doping with O/N/B generates a greater number of active centers on adjacent carbon atoms than doping with O and O/N atoms. As a result of synergy with enhanced wetting ability (contact angle of 9.26°) derived from amino groups and hierarchical porous structure, ON-CA has the most optimized adsorption capacity (-1.62 eV) and diffusion barrier (0.12 eV) of K^+ . The optimal pathway of K^+ in ON-CA is along the carbon ring with N or O doping. As K^+ storage material for supercapacitors and ion batteries, it shows an outstanding specific capacity and capacitance, electrochemical stability, and rate performance. Especially, the assembled symmetrical K^+ supercapacitor demonstrates an energy density of 51.8 Wh kg^{-1} , an ultrahigh power density of 443 W kg^{-1} , and outstanding cycling stability (maintaining 83.3% after 10,000 cycles in 1 M KPF₆ organic electrolyte). This research provides valuable insights into the design of high-performance potassium ion storage materials.

Kaijun Xie, Xin Liu, and Haolin Li contributed equally to this work.

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agarose, carbon aerogels, O/N/B codoping, potassium-ion battery, potassium-ion supercapacitor

1 | INTRODUCTION

Three-dimensional (3D) carbon aerogel (CA) is widely used in catalysis, adsorption, separation, energy storage, biomedicine, and other fields due to their excellent electrical conductivity, sizable specific surface area, high-temperature stability, low density, high porosity, and biocompatibility.^{1,2} Specifically, their 3D interconnected networks with open pores go about as repositories of electrolytes and give low-obstruction pathways for mass transfer, which makes them more promising applications in electrochemical energy storage.^{3,4} However, the CA are typically fabricated using organic materials such as phenolic resins and polyacrylonitrile, which are usually derived from petroleum derivatives.⁵ The process is well known to be both nonrenewable and environmentally harmful.⁶ Therefore, finding a way with low-cost, renewable, and environmental precursors to prepare CA is significant.^{7,8}

The utilization of biopolymers extracted from biomass (e.g., bamboo, husk, proteins) as precursor materials to synthesize 3D porous CA has garnered extensive research interest.^{9,10} The ample availability, economic viability, nontoxicity, and environmental compatibility of these materials meet the essential prerequisites for their employment as electrode components in diverse electrochemical applications.^{11–13} However, when using biomass-derived CA as electrode materials for ion batteries and supercapacitors, the pore collapse and pore size reduction lead to a diminished cycling lifespan.¹⁴ In addition, the presence of considerable interstitial voids impedes the mass transfer process.^{15–17} In order to improve the freestanding electrode material performance, heteroatom (N, O, or B) doping has been generally employed.^{18,19} It has been proven that doping carbonaceous materials with heteroatoms results in an improvement in various properties, including enhancement in the conductivity and tunable interlayer distance of carbon.^{20–23} Among them, N-doping induces changes to the electronic structure of CA,²⁴ leading to an increase in surface chemical activity.²⁵ O-doping alters the electronic density and energy level structure of CA, thereby facilitating electron transfer and conduction.^{26,27} B-doping results in changes to the hybridization of carbon atoms in sp^3 orbitals shifting toward sp^2 orbitals and a consequent increase in the hole concentration of carbon materials.^{28,29} However, the synergistic effect of codoping

(N, O, and B) in biomass-derived CA is yet to be investigated.^{30,31} Furthermore, the effect of O/N/B codoping on the potassium ion (K^+) deintercalation pathway remains unclear.^{32,33}

Herein, we have synthesized O/N/B functionalized CA (ONB-CA) derived from agarose, which has a huge specific surface area of $409.62 \text{ m}^2 \text{ g}^{-1}$ and a hierarchical porous structure for ON-CA. As potassium ion energy storage material, ON-CA achieves an ultrahigh specific capacitance of 385 F g^{-1} at a current density of 0.5 A g^{-1} in electrochemical double-layer supercapacitors with 6 M KOH electrolyte and maintains 93.5% of its capacitance after 10,000 cycles. Symmetric potassium ion supercapacitors assembled with ON-CA exhibit an energy density of 51.8 Wh kg^{-1} , an ultrahigh power density of 443 W kg^{-1} , and excellent cycling stability (maintaining 83.3% after 10,000 cycles in 1 M KPF_6 organic electrolyte). Furthermore, the assembled potassium-ion battery (PIB) with ON-CA exhibits a specific capacity of 174 mAh g^{-1} and maintains 105 mAh g^{-1} after 300 charge–discharge cycles. Density functional theory (DFT) calculations were performed to explore the O/N/B functionalization and K^+ storage mechanism. The synergistic effect of enhanced wetting ability (contact angle of 9.26°) and ion conductivity (increased electronic states near the Fermi level) in ON-CA enhances K^+ adsorption energy (-1.62 eV) and lowers the energy barrier of K^+ diffusion (0.12 eV). In addition, the optimal pathway of K^+ in ON-CA is along the carbon ring with N or O doping. This work proposes an innovative strategy for the widespread application of CAs in environmentally friendly and efficient energy storage devices.

2 | EXPERIMENTAL SECTIONS

2.1 | Material preparation

Agarose was obtained from Aladdin, and other chemicals were purchased from Sinopharm Holding Chemical Reagent Co., Ltd. All chemicals were used as received. First, agarose (6 g) and deionized water (300 g) were heated in a water bath at 90°C while stirring. Water was added at least three times during this process, maintaining the liquid level close to 300 mL. Subsequently, the dissolved agarose solution was poured preformed molds

and subjected to freeze-drying. The freeze-drying process is described as follows: The hydrogel samples were stored at -18°C for 24 h in an ultra-low temperature refrigerator (TF-60-400-LA). Subsequently, the samples were subjected to freeze-drying in a freeze dryer (LGJ-10N) maintained at a temperature of -35°C and a pressure of 0.0 Pa for a duration of 48 h to obtain the agarose aerogels, labeled as O-AA.

Similarly, agarose (6 g) and deionized water (300 g) were heated in a water bath at 90°C while stirring. Water was added at least three times during this process, maintaining the liquid level close to 300 mL. Cyanamide was added, and the mass ratio of the added amount to agarose was expanded according to 2:1, 1:1, 1:2, 1:3. Finally, the same posttreatment method as the agarose was used to obtain N-doped agarose aerogels, recorded as ON-AA.

To obtain ONB-AA, agarose (6 g) and deionized water (300 g) were heated in a water bath at 90°C while stirring. Water was added at least three times during this process, maintaining the liquid level close to 300 mL. Tetrabutylammonium borohydride was added, and the mass ratio of the added amount to agarose was expanded by 3:1, 2:1, 1:1, 1:2. Finally, the same post-treatment method was used to obtain ONB-AA.

Thermal decomposition of O-AA, ON-AA, and ONB-AA was carried out in a homemade tubular furnace. Thermogravimetric analysis (TGA) was conducted for the O-AA, ON-AA, and ONB-AA under the N_2 atmosphere to give information on the calcination temperature (Figure S1). From 279°C to 800°C , weight losses of 54%, 55%, and 57% are observed, which might be due to organics in the sample being oxidized during pyrolysis to produce volatile gases (CO , CO_2 , and H_2O) and carbonaceous residues. Based on the analysis conducted using thermogravimetric, the temperature of 800°C was chosen for the carbonization process, and three sets of samples were prepared using CA. The samples underwent controlled heating with a temperature ramp of $4^{\circ}\text{C min}^{-1}$ until reaching 800°C , following which they were subjected to a 2-h pyrolysis process under an inert N_2 . The resulting products were designated as O-CA, ON-CA, and ONB-CA, correspondingly. Based on a meticulous examination of the pertinent literature and comprehensive data analysis, the densities of O-CA, ON-CA, and ONB-CA have been determined to be 0.17, 0.21, and 0.26 g cm^{-3} , respectively.

2.2 | Material characterization

Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) were utilized to

investigate the morphology of O-CA, ON-CA, and ONB-CA. The GeminiSEM 500 and JEOL 2100F instruments were employed for SEM and TEM analysis, respectively. To determine the crystal properties of the sample, an X-ray diffractometer (XRD) equipped with $\text{Cu K}\alpha$ radiation was employed. The XRD instrument used in this study was the Rigaku Ultima IV. Raman spectra were acquired at a wavelength of 532 nm using a DXR 2xi spectrometer. The specific surface area of the sample was determined utilizing the nitrogen adsorption/desorption Barrett-Emmett-Teller (BET) technique. The BET analysis was conducted using an ASAP2460 instrument. The pore size distribution of the sample was quantified using the Barrett-Joyner-Halenda (BJH) method. Data were collected using a monochromatic ($\text{K}\alpha$ radiation) source on an ESCALAB 250Xi spectrometer. To observe the hydrophilicity of the sample thin film surface, contact angle measurements were performed. All measurements were conducted at room temperature using deionized water. Approximately $10 \pm 2 \mu\text{L min}^{-1}$ of water droplets were added onto the thin film surface. Contact angle measurements were performed using a contact angle measurement instrument equipped with a CCD camera. The specific instrument used in this study was the model provided by Theat Co., Biolin. TGA was performed on the sample in N_2 at $30\text{--}800^{\circ}\text{C}$ using a NETZSCH STA 449F3 thermogravimetric analyzer.

2.3 | Electrochemical characterization

A nitrogen methyl pyrrolidone solution was incorporated into and thoroughly mixed with 80 wt% of O-CA, ON-CA (cyanamide: agarose = 1:2 is the best), and ONB-CA (agarose:tetrabutylammonium borohydride = 2:1 is the best), 10 wt% of carbon black, and 10 wt% of polytetrafluoroethylene, to prepare the electrode. The ensuing composite was meticulously deposited onto $1 \text{ cm} \times 1 \text{ cm}$ foamed nickel substrate and subsequently desiccated in a temperature-controlled oven. For the purpose of electrochemical evaluation, a three-electrode configuration was employed, wherein the nickel foam, Hg/HgO electrode, and platinum sheet laden with electrode materials served as the working, reference, and counter electrodes, respectively, all immersed in a 6 M KOH aqueous electrolyte solution. The electrochemical performance of the single electrode was examined by recording cyclic voltammetry (CV) curves at various scan speeds within the voltage range of -1 to 0 V . Galvanostatic charge-discharge

(GCD) was seen in the current density range of 0.5–20 A g⁻¹ when constant current charge and discharge experiments carried out at the same voltage window. Electrochemical impedance spectroscopy (EIS) was performed at an open circuit potential of 5 mV and a frequency range of 0.01 Hz to 100 kHz, and the Nyquist plot was analyzed. Two electrode materials of identical quality were meticulously chosen and segregated by a glass fiber separator to fabricate a symmetrical supercapacitor, which was subsequently immersed in a 1 M KPF₆ organic electrolyte solution for electrochemical characterization. All electrochemical experiments were conducted at ambient temperature utilizing an electrochemical workstation. (CHI 760E; CH Instruments Inc.).

The O-CA, ON-CA, and ONB-CA were blended with conductive carbon black and polyvinylidene fluoride in a weight ratio of 8:1:1, utilizing nitrogen-methyl pyrrolidone as the solvent. Subsequently, the resulting slurry was meticulously applied onto a copper foil, transferred to a vacuum, and subjected to drying at 120°C for a duration of 12 h. The dried slurry was further sectioned into circular disks, possessing a diameter of 1.0 cm and a loading weight of 0.8–0.9 mg. The disks were then meticulously assembled with a potassium metal sheet, serving as the counter electrode, within a glove box under argon protection, and subsequently fashioned into CR2016 half-cells. The electrolyte utilized in the cells was 1 M KPF₆ carbonate ester/dimethyl carbonate (1:1, in wt%). The cycle stability and rate performance were rigorously evaluated using the Land CT2001A battery testing system, within a voltage range of 0.01–2.6 V. The CV profiles were obtained using an electrochemical workstation (CHI 760E).

The gravimetric specific capacitance (C) was meticulously computed utilizing the following equations³⁴:

$$C = \frac{I\Delta t}{m\Delta V}, \quad (1)$$

$$C = \frac{\int IdV}{vmV}, \quad (2)$$

where I is the current density, ΔV is the potential change within the discharge time Δt , V is potential, v is the potential scan rate, and m is the mass of the electroactive materials (capacitance unit: F g⁻¹).

The gravimetric specific capacity, energy density (E), and power density (P) of the symmetrical system were rigorously calculated utilizing the following formula³⁵:

$$E = \frac{CV^2}{2} = \frac{C_p\Delta U^2}{7.2}, \quad (3)$$

$$P = \frac{E}{\Delta t} = \frac{E \times 3600}{\Delta t}. \quad (4)$$

The aforementioned formula incorporates the variables I , V , v , m , and Δt , which, respectively, represent the current density, working range, scan rate, weight of the active materials of the two electrodes, and discharge time (s). The units of energy density and power density are expressed in Wh kg⁻¹ and W kg⁻¹, respectively.

2.4 | Electrochemical kinetic analysis

Throughout the CV test, the peak current at varying scan rates was meticulously recorded, and the correlation between the scan rate and the peak current was analyzed to differentiate between diffusion and pseudocapacitive behavior during charge and discharge. This relationship can be mathematically expressed by the following empirical formula³⁶:

$$\log i(v) = \log a + b \log v. \quad (5)$$

Both a and b are variable parameters in the aforementioned formula. It is pertinent to note that the b value corresponds to the slope of the $\log i$ versus $\log v$ line.

When $b = 0.5$, the peak current varies as a function of 0.5 of the scan rate. Using Faraday intercalation, conventional diffusion controls the charge transfer process.

When $b = 1$, the peak current fluctuates straightly with the sweep rate, and the energy stockpiling process relies upon the adsorption/desorption of surface charges.

When $b = 0.5$ –1, this is a transitional region between pseudocapacitive materials and battery-type materials.

Upon conducting a comprehensive CV test, and considering all relevant factors, a functional relationship between current and scan rate can be rigorously derived as follows³⁷:

$$i(V)/v^{1/2} = k_1 v^{1/2} + k_2. \quad (6)$$

The values of k_1 and k_2 can be meticulously calculated by obtaining the slope and intercept of the linear plot of $i(V)/v^{1/2}$ and $v^{1/2}$, respectively.

2.5 | DFT calculations

The Vienna ab initio simulation Package was utilized to conduct DFT calculations. The calculations

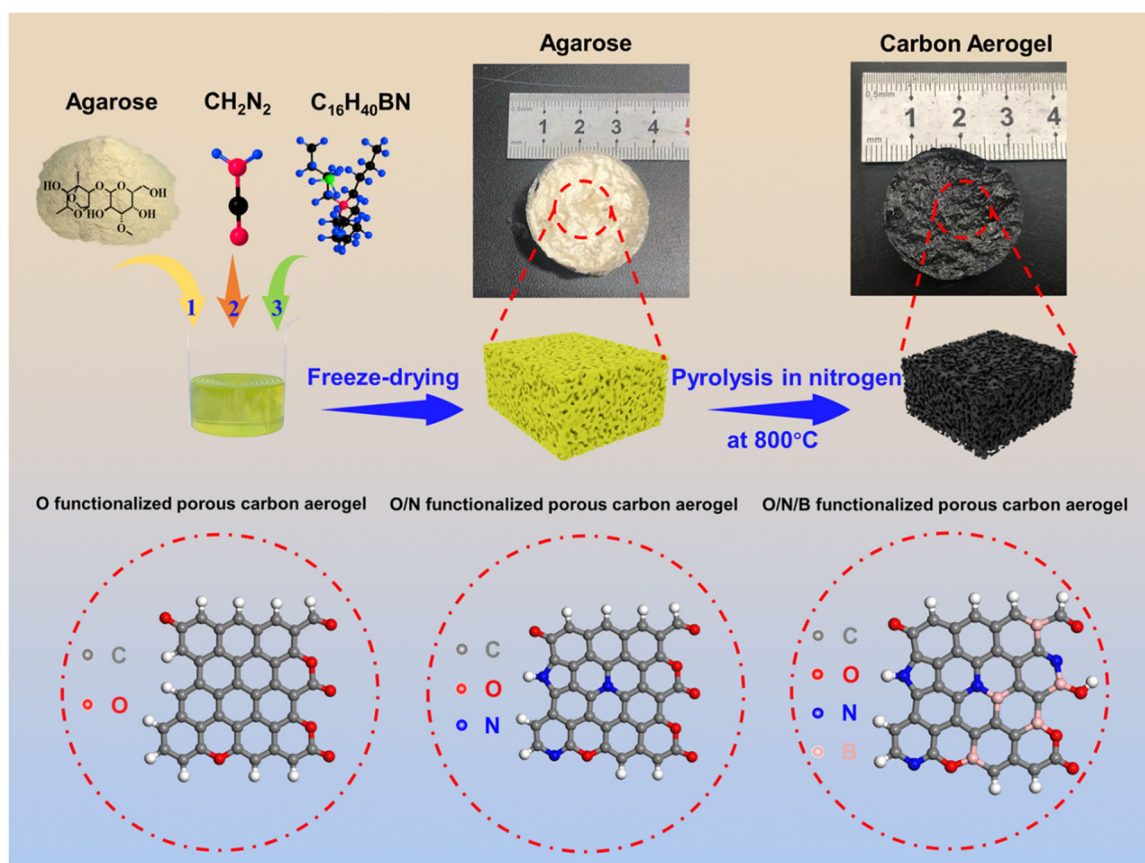
employed a plane wave basis set with an energy cutoff of 500 eV, projector augmented wave pseudopotentials, and the generalized gradient approximation parameterized by Perdew, Burke, and Ernzerhof for exchange-correlation functional.³⁸ To account for the dispersion correction, Grimme's semiempirical DFT-D3 scheme was adopted.³⁹ The vacuum space was established as 15 Å for all three directions in this study. The model structures underwent complete optimization for both ionic and electronic degrees of freedom, employing convergence criteria of 10^{−4} eV for electronic energy and 10^{−2} eV/Å for the forces acting on each atom. The binding energy was defined as⁴⁰

$$E_b = E_{K/surf} - E_{surf} - E_K. \quad (7)$$

The binding energy was determined by computing the total energies of the heteroatom-doped graphene with and without the adsorption of a K atom, denoted as $E_{K/surf}$ and E_{surf} , respectively. Here, E_K represents the energy of a single K atom in the bulk position.

3 | RESULTS AND DISCUSSION

Scheme 1 illustrates that the synthesis of O-CA entailed dissolving 2% agarose in hot water at 90°C, which was subsequently cooled to room temperature to form a hydrogel. The formation of the resulting hydrogel is attributed to the intermolecular hydrogen bonding and van der Waals interactions between the agarose molecules. Subsequently, the hydrogel was freeze-dried to produce agarose aerogel, which was then pyrolyzed at high temperatures under the protection of N₂. During the pyrolysis process, the water molecules in the agarose polysaccharide were removed, and the chemical bonds within the molecules began to break, leading to the production of carbides. Similarly, in the preparation of ON-CA, cyanamide molecules underwent polyaddition and polycondensation reactions with increasing temperature, resulting in specific reactions (Figure S2).^{41,42} At a temperature of 520°C, melem underwent polycondensation to form carbon nitride. At an elevated temperature of 800°C, the process of nitrogen-carbon decomposition yields a substantial production of NH₃. The aforementioned gases serve as agents for forming



SCHEME 1 Schematic diagram of O-CA, ON-CA and ONB-CA preparation. CA, carbon aerogel.

pores and as dopants of nitrogen in the preparation of N, O-codoped CA. In the preparation of ONB-CA, tetrabutylammonium borohydride dissociated in water to produce tetrabutylammonium ions and boron hydroxide ions, while agarose formed a colloidal state in water. During high-temperature pyrolysis, the reaction between B and N₂ with carbon chains leads to the formation of H₂ and BN. Meanwhile, through decomposition and carbon-chain formation, tetrabutylammonium is utilized to synthesize O/N/B doped CA, as depicted in Figure S3.⁴³

The crystal structure of O-CA, ON-CA, and ONB-CA was analyzed using XRD spectroscopy (Figure 1A). All samples show two prominent diffraction peaks at 24° and 44°, relating to (002) and (101) crystal planes of carbon, respectively. In comparison to O-CA and ONB-CA, the (002) peak of ON-CA exhibits greater width, which signifies a lower degree of crystallinity in ON-CA. The Raman spectra of all samples exhibit two prominent peaks

at 1336 and 1575 cm⁻¹, which correspond to the D-band (associated with sp³ defective carbon induced by disorder) and G-band (associated with sp² bonded graphitic carbon), respectively.⁴⁴ As shown in Figure 1B, the D-band and G-band intensity ratios (I_D/I_G) are 0.82, 0.85, and 0.84 for O-CA, ON-CA, and ONB-CA, respectively. The increased disorder degree in ON-CA compared with those in O-CA and ONB-CA is owing to the introduction of N, O codoping into the CA framework, creating more defects and bulges on the surface of CA.

The specific surface area and pore size of O-CA, ON-CA, and ONB-CA were ascertained by means of N₂ adsorption-desorption measurements carried out using BET and BJH techniques (as illustrated in Figure 1C,D and Table S1). The adsorption-desorption isotherm of O-CA displays a type IV isotherm, whereas ON-CA and ONB-CA exhibit a combination of type I and type IV isotherms, indicating the coexistence of micropores and

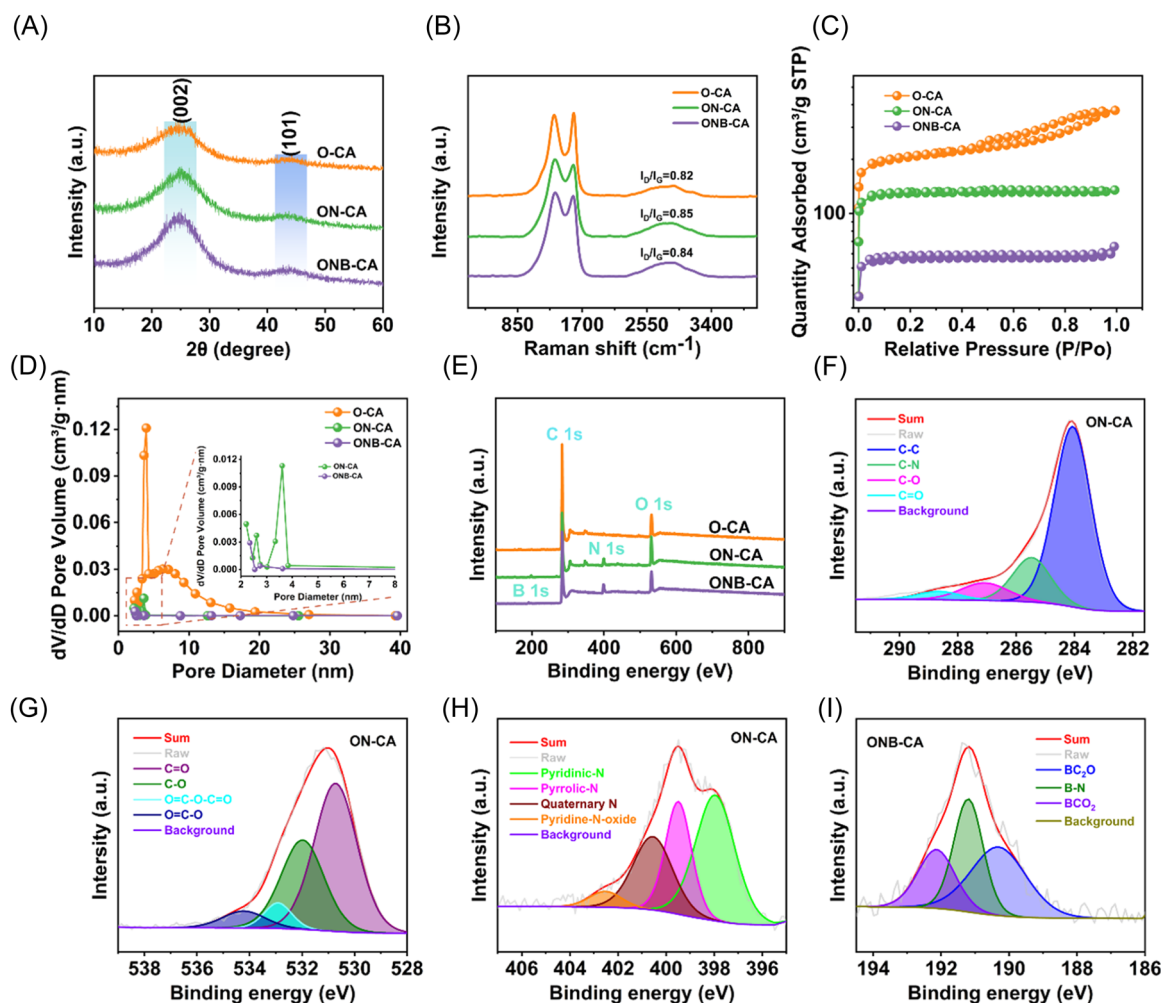


FIGURE 1 (A) XRD patterns of O-CA, ON-CA, and ONB-CA. (B) Raman spectrum of O-CA, ON-CA, and ONB-CA. (C) The isotherms of N₂ adsorption/desorption for O-CA, ON-CA, and ONB-CA. (D) The distribution of pore sizes for O-CA, ON-CA, and ONB-CA. (E) XPS survey spectrum of O-CA, ON-CA, and ONB-CA. High-resolution spectra of (F) C 1s, (G) O 1s, and (H) N 1s for ON-CA, and (I) B 1s for ONB-CA.

mesopores.⁴⁵ The difference in structure results in specific surface areas of 669.14, 409.62, and 179.88 m² g⁻¹ for O-CA, ON-CA, and ONB-CA, respectively. The reduction in specific surface area can be ascribed to the formation of organic gas during the high-temperature pyrolysis process of CA, which causes the etching of the CA material and consequent disruption of the pore structure. Furthermore, O-CA also exhibits a hysteresis loop (typical type H4 loop), which is primarily attributed to the existence of large mesopores. However, the N₂ adsorption isotherm of ON-CA and ONB-CA does not exhibit a hysteresis loop even at higher pressure, indicating the presence of only a few mesopores. According to the pore size distribution diagram (Figure 1D), both ON-CA and ONB-CA exhibit a relatively concentrated pore size distribution (below 4 nm), whereas O-CA demonstrates a dispersed distribution (ranging from 1 to 20 nm). This observation is consistent with the N₂ adsorption/desorption isotherms. Based on the pore size distribution (Figure 1D), ON-CA and ONB-CA exhibit a relatively centralized pore size distribution (below 4 nm), whereas O-CA displays a decentralized distribution (ranging from 1 to 20 nm), which is consistent with the N₂ adsorption/desorption isotherms.

Furthermore, XPS was utilized to investigate the surface functional groups and chemical bonding states of CA. The complete spectra shown in Figure 1E demonstrate the presence of O, N, and B elements in CA, with the elemental composition of each sample summarized in

Table S2. Figure S4 illustrates the C 1s and O 1s spectra of O-AC. The peaks observed in the high-resolution C 1s spectrum of ON-CA (Figure 1F) can be attributed to C-C (284.07 eV), C-N (285.39 eV), C-O (287.12 eV), and C=O (288.79 eV) functionalities. The O 1s spectrum (Figure 1G) reveals the existence of functional groups containing oxygen, including C=O (530.74 eV), C-O (531.99 eV), O=C-O-C=O (532.94 eV), and O=C-O (534.24 eV). The aforementioned oxygen functionalities undergo redox reactions during the charge and discharge processes, thereby imparting additional pseudocapacitance to the material. The N 1s spectrum (Figure 1H) exhibits the presence of pyridinic-N (398.54 eV), pyrrolic-N (400.03 eV), N-O (401.79 eV), and N-Q (401.14 eV).⁴⁶ Pyridinic-N and pyrrolic-N are known to create defects and furnish electrically active sites, which aid in the storage of potassium ions. Quaternary amine nitrogen is in favor to increase the wettability of CA. The forms of C 1s (Figure S5A), N 1s (Figure S5B), O 1s (Figure S5C), and B 1s (Figure 1I) in ONB-CA were identified as B-N (191.2 eV), BCO₂ (192.2 eV), and BC₂O (190.6 eV).⁴⁷ The nonpolar functional groups of BCO₂ and BC₂O are insoluble in water, leading to decreased hydrophilicity of ONB-CA. Thus, contact angles were investigated to verify the hydrophilicity of O-CA, ON-CA, and ONB-CA (Figure S6), which are 27.61°, 9.26°, and 37.03°, respectively. Notably, ON-CA demonstrated exceptional hydrophilicity, which can be attributed to the copious presence of hydrophilic functional groups containing nitrogen on its surface. This will effectively promote the adsorption of

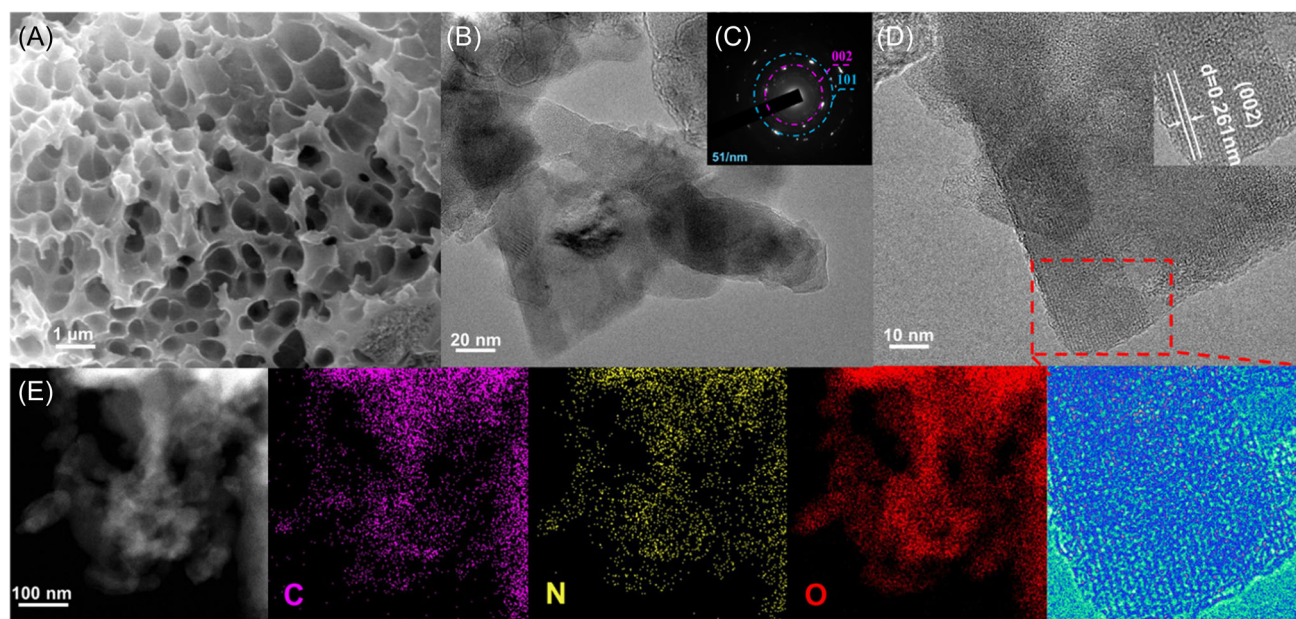


FIGURE 2 (A) SEM images of ON-CA. (B, D) HRTEM images of ON-CA. (C) Corresponding fast fourier transformation pattern of (B). (E) Energy dispersive X-ray spectrometry elemental mapping images of C, N, and O in ON-CA, respectively.

charged particles from the electrolyte onto the charged adsorbent electrode, thereby amplifying the energy storage of potassium ions.

The morphologies of O-CA, ON-CA, and ONB-CA were examined by means of SEM. As shown in Figure 2A, ON-CA exhibits a 3D macroporous honeycomb structure with varying pore sizes, which likely attributes to the decomposition of cyanamide into a significant amount of ammonia gas during the high-temperature pyrolysis process. Ammonia acts as a pore-forming agent for the carbon, resulting in the formation of macropores.⁴⁸ Furthermore, micropores and mesopores in carbon can also arise from phase separation between hydrophobic carbon and water during pyrolysis.⁴⁹ O-CA exhibits similar morphology with ON-CA (Figures S7 and S8).

In summary, the BET and pore size distribution results indicate that the hierarchical porous structure consisting of micro, meso, and macropores offers efficient pathways for the rapid transport and mutual penetration of electrolyte ions. Additionally, these structures reduce the diffusion resistance of ions, thereby promoting energy storage.⁵⁰ The pore channels of ONB-CA (Figure S9) collapse, which results in a reduction in pore size or partial pore blockage. High-resolution TEM (HRTEM) was used for further study of the morphology of ON-CA (Figure 2B–D). Rich micropores and a small amount of crystal type can be observed in the image, with a d -spacing of 0.261 nm corresponding to the (002) lattice fringe of carbon in the XRD results. The final image in

Figure 2E is an enlarged portion of Figure 2D, better reflecting the distribution of the lattice fringe. The elemental mapping image (Figure 2E) shows that C, O, and N were uniformly dispersed on ON-CA, further demonstrating the codoping of N and O in CA.

The electrochemical performance of O-CA, ON-CA, and ONB-CA for potassium ion storage was evaluated using a three-electrode supercapacitor system with a 6 M KOH electrolyte.⁵¹ The CV curves of O-CA, ON-CA, and ONB-CA were estimated at a scan rate of 100 mV s^{-1} , as shown in Figure 3A. Figure 3B displays the GCD curves for each sample as determined at a current density of 1 A g^{-1} . The results indicate that the ON-CA electrode demonstrates superior electrochemical performance, as evidenced by its longer discharge time compared to the other electrodes. It is evident that the ON-CA electrode exhibits a higher oxidation–reduction peak and CV curve area compared to the other control samples. These findings suggest that the ON-CA electrode possesses a higher specific capacitance and faster redox reaction rate. Figure 3C presents the specific capacitance of each electrode, as determined by the GCD data. The mass-specific capacitance of the ON-CA electrode was determined to be 385.3 F g^{-1} at a current density of 0.5 A g^{-1} , which is notably higher than those of O-CA and ONB-CA. The data presented in Figure 3D illustrates that the specific capacitance of ON-CA surpasses that reported in all previous literature (Table S3), suggesting a superior potassium storage property. The distinctive 3D large-pore honeycomb structure of ON-CA offers a significant

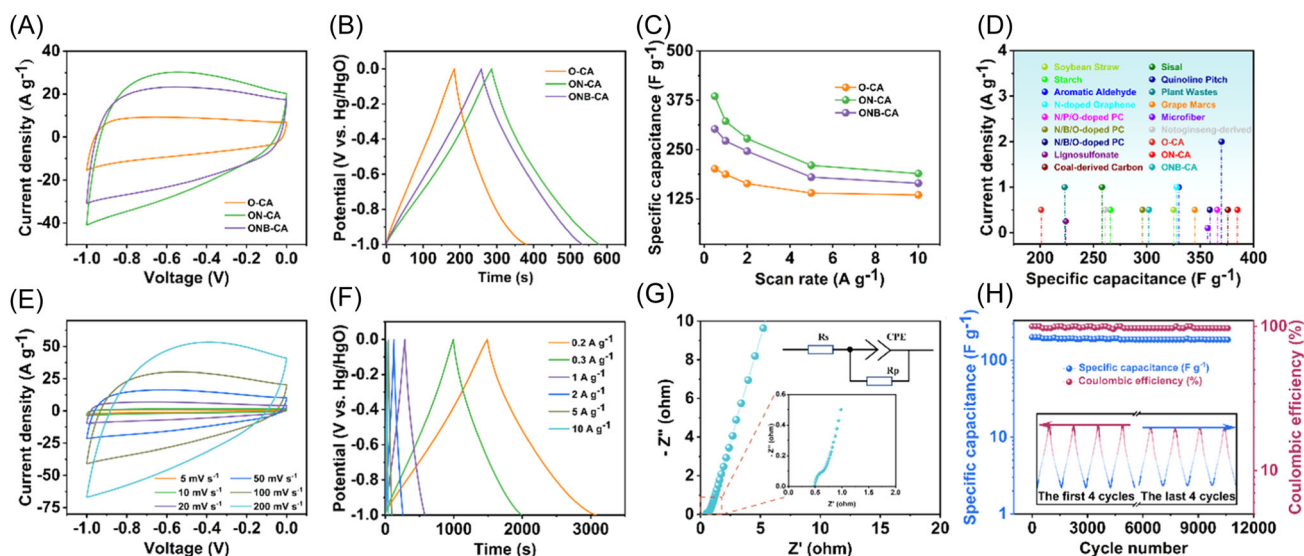


FIGURE 3 (A) O-CA, ON-CA, and ONB-CA CV curves in 6.0 M KOH aqueous solution at a scan rate of 100 mV s^{-1} . (B) O-CA, ON-CA, and ONB-CA CV curves at 0.5 A g^{-1} in current density. (C) O-CA, ON-CA, and ONB-CA specific capacitances at various current densities. (D) Comparison of the specific capacitance of ON-CA with other materials.^{52–66} (E) CV curves of ON-CA in 6.0 M KOH aqueous solution at various scan rates of 5, 10, 20, 50, 100, and 200 mV s^{-1} . (F) ON-CA charge–discharge curves at different current densities, measured galvanostatically. (G) Nyquist plots of ON-CA electrodes. (H) 10,000 cycles of testing at 5 A g^{-1} of the electrochemical stability of ON-CA.

number of active sites and ample surface area, thereby facilitating ion/electron transfer.

Figure 3E depicts the CV curves of the ON-CA electrode material, which were obtained by testing at various scan rates within the voltage range of -1 to 0 V. It was observed that the shape of the CV curve remains largely unchanged as the scan rate increases from 5 to 200 mV s^{-1} , suggesting that the ON-CA electrode exhibits favorable electrochemical stability, exceptional electrochemical reversibility, and swift ion/electron conductivity. The observed phenomenon can be ascribed to the reversible redox reaction, which is induced by the codoping of N and O functional groups, as depicted in Figure S10. The optimized CV and GCD data for ON-CA are presented in Figures S11–S14. The electrochemical performance of O-CA and ONB-CA electrode materials was evaluated under different scanning rates, as illustrated in the CV curves presented in Figure S15. Meanwhile, the CV and GCD performance of ONB-CA under optimized conditions is presented in Figures S16–S19. The GCD curves of ON-CA, presented in Figure 3F, exhibit distinct voltage plateaus and symmetric profiles, which are indicative of the electrode's high Coulombic efficiency and redox reversibility. The GCD curves of O-CA and ONB-CA at various current densities are shown in Figure S20.

Moreover, EIS was performed on ON-CA and the reference electrode within the frequency range of 10^{-2} – 10^6 Hz, as illustrated in Figure 3G. The R_s values of each electrode were determined by magnifying the high-frequency region of the plot.⁶⁷ Comparing the internal resistance of the three electrode samples, ON-CA exhibits the lowest value (0.49Ω), indicating the fastest ion conduction among the samples (Figure S21). Based on the impedance plot of the ON-CA electrode, it can be inferred that the electrode surface in contact with the KOH electrolyte exhibits a low charge transfer resistance (R_p). Moreover, the steep low-frequency slope in the impedance plot of the ON-CA electrode indicates a low diffusion resistance during Faradaic redox processes. The impedance plot's equivalent circuit diagram is illustrated in Figure 3G, where R_s represents the internal resistance between the electrode and electrolyte, R_p represents the charge transfer resistance, and constant phase angle element represents the Warburg resistance. Furthermore, the 3D interconnected skeleton of the ON-CA electrode offers superior mechanical stability and prolonged durability. As shown in Figure 3H, even after 10,000 cycles at a current density of 20 A g^{-1} , the ON-CA electrode maintains a capacitance retention of 92.5%, while O-CA and ONB-CA exhibit retention rates of 90.5% and 81.2%, respectively (Figure S22).

To enhance the comprehension of the electrochemical reaction rates, diffusion, and pseudocapacitive behavior of ONB-CA and the control samples during the process of charge and discharge, the energy storage pathways were investigated. To evaluate the CV curves of the control samples, CV measurements were performed at various scan rates ranging from 5 to 100 mV s^{-1} . The CV curves of O-CA, ON-CA, and ONB-CA are presented in Figure S23. Figure 4A–C presents the calculated oxidation peak b values (Equation 5) of the samples as 0.72 , 0.67 , and 0.76 , representing the transition region between pseudocapacitive and battery-type electrode materials. The energy storage mechanisms of electrode materials are primarily predicated on the ensuing pathways: During the ion intercalation process, the Faradaic contribution arises from the Faradaic effect generated by the diffusion of surface atomic charges (commonly referred to as pseudocapacitance), while the non-Faradaic contribution stems from the double-layer effect. Figure 4D–F presents the calculated pseudocapacitance contribution rates (Equation 6) of O-CA, ON-CA, and ONB-CA at a scanning rate of 50 mV s^{-1} . The pseudocapacitance contribution rates of O-CA, ON-CA, and ONB-CA are 75.59%, 81.02%, and 80.26%, respectively. The maximum specific surface area of O-CA, however, has the minimum capacitance contribution, owing to the small influence of oxides on the surface wettability and the relatively low charge storage density. An excess of oxygen atoms can affect the conductivity of the electrode material. The maximum capacitance contribution of ON-CA is mainly due to the strong electronegativity of nitrogen and oxygen atoms and hydrophilic functional groups such as hydroxyl and amino formed by them, which enhance the wettability of the material with the electrolyte solution. ONB-CA contains many heterogeneous atoms, but its capacitance contribution is lower than that of ON-CA, mainly because oxygen and nitrogen are two very effective doping elements that can introduce electrons or holes and increase the material capacitance contribution. Boron has a small atomic radius and is difficult to dope into the carbon network of CAs, thus having a limited effect on improving capacitance performance.⁶⁸ Furthermore, the pseudocapacitive and diffusion currents of O-CA, ON-CA, and ONB-CA at various scan rates are depicted in Figure 4G–I which illustrates that the pseudocapacitive contribution to the capacitance increases with the increasing scan rate due to the rapid redox reactions at high scan rates.

To enhance the comprehension of the electrode energy storage capabilities of CA, symmetric supercapacitors were constructed utilizing a 1 M KPF_6 organic electrolyte.⁶⁹ As illustrated in Figure 5A, when subjected

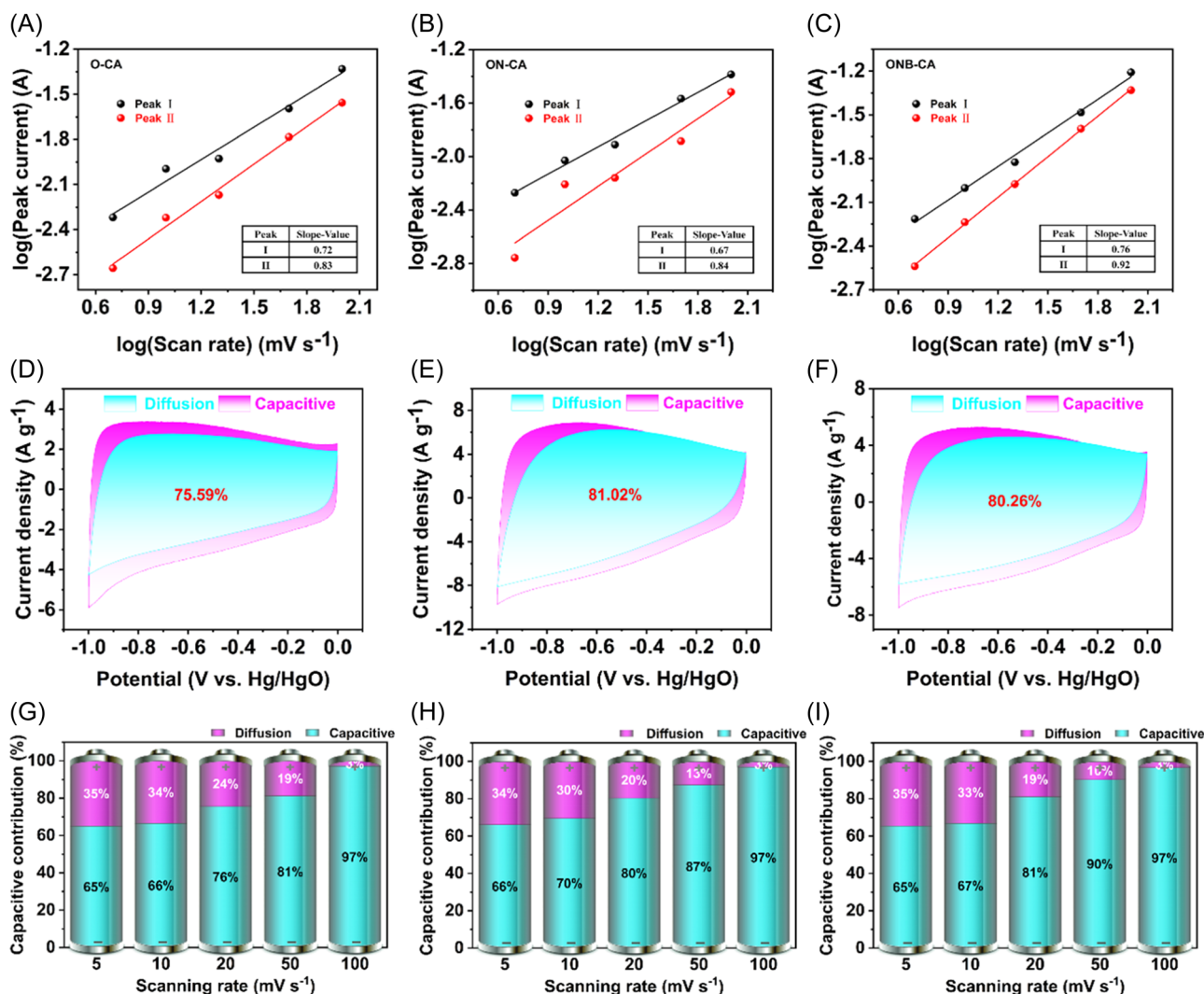


FIGURE 4 The dependence of parameter b for (A) O-CA, (B) ON-CA, and (C) ONB-CA. The separation of capacitive and diffusion contributions for (D) O-CA, (E) ON-CA, and (F) ONB-CA at 20 mV s⁻¹. Capacitance contribution rates of (G) O-CA, (H) ON-CA, and (I) ONB-CA electrodes at diverse scan rates.

to a scan rate of 50 mV s⁻¹, it is apparent that the CV curve area of ON-CA is greater than that of O-CA and ONB-CA, indicating a notable weak redox peak arising from the reversible oxidation–reduction reactions of the nitrogen–oxygen functional groups. Figure 5B portrays the GCD curve at a current density of 0.5 A g⁻¹. The nearly symmetrical linear curves of all three devices suggest excellent capacitive performance. The GCD data analysis reveals the specific capacitances computed at varying power densities, as presented in Figure 5C. At a current density of 0.5 A g⁻¹, the specific capacitance is determined to be 102.3 F g⁻¹, which is significantly greater than those of O-CA and ONB-CA, as illustrated in Figure 5D. When reaching a power density of 443 W kg⁻¹, the energy density of ON-CA can attain 51.8 Wh kg⁻¹, exceeding the energy density of other carbon-based symmetric supercapacitors reported in the

literature (Table S4). The energy densities of O-CA and ONB-CA are 43.8 and 16.3 Wh kg⁻¹, respectively.

As shown in Figure 5E, the CV curve exhibits a well-defined rectangle from 0 to 3.0 V. Beyond 3.0 V, the curve shape distorts, leading to anodic current. Hence, the working voltage range is restricted to 0–2.8 V. The working voltage range of O-CA and ONB-CA is also limited to 0–2.8 V (Figure S24). Figure 5F presents the CV curves at varying scan rates, illustrating that the rectangular region progressively expands in correspondence with the scan rate and retains an optimal rectangular morphology. This observation signifies the electrolyte's stability within the specified voltage range. The CV curves of O-CA and ONB-CA at different scan rates are shown in Figure S25. Figure 5G displays the GCD curves of ON-CA at various current densities, demonstrating a near-symmetrical linear response. This

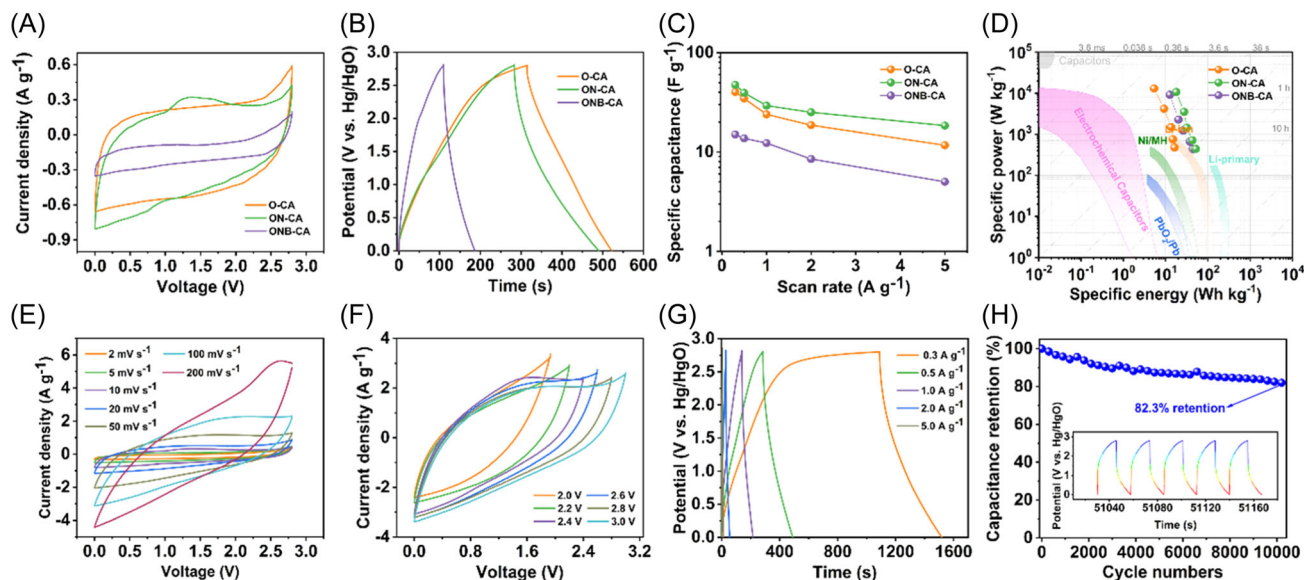


FIGURE 5 (A) O-CA, ON-CA, and ONB-CA CV curves at a scan rate of 50 mV s^{-1} . (B) O-CA, ON-CA, and ONB-CA CV curves with a current density of 0.5 A g^{-1} . (C) O-CA, ON-CA, and ONB-CA specific capacitances at various current densities. (D) Ragone plot of O-CA, ON-CA, and ONB-CA symmetric supercapacitor. (E) ON-CA//ON-CA CV curves symmetrical supercapacitor with a scan rate of 100 mV s^{-1} and different operating voltages. (F) CV curves of an ON-CA//ON-CA symmetrical supercapacitor at different scan speeds in the voltage range of 0–2.8 V. (G) GCD curves of the ON-CA//ON-CA symmetric supercapacitor at various current densities. (H) Electrochemical stability of the ON-CA symmetric supercapacitor 10,000 cycles test.

characteristic signifies outstanding capacitive performance and exceptional electrochemical reversibility. Figure S26 presents the GCD curves of O-CA and ONB-CA at distinct current densities. Figure 5H depicts the electrochemical cycling of ON-CA in a 1 M KPF₆ electrolyte for 10,000 cycles, with the corresponding specific capacitance remaining at 83.3% of the initial value. This indicates an increased wettability of the ON-CA material, facilitating the entry of electrolyte ions into the interlayer of CA to store more charge, resulting in an increase in specific capacitance with cycling. The long-term retention rates of O-CA and ONB-CA are 81.3% and 76.6%, respectively (Figure S27). Two-electrode Nyquist plots of O-CA, ON-CA, and ONB-CA were conducted before and after cycling (Figure S28). The test results reveal a slight increase in internal resistance, and all samples exhibit a minor increase in R_s following the cycling tests (initial and cycled R_s values are 72 and 75 Ω for O-CA, 66 and 68 Ω for ON-CA, and 76 and 79 Ω for ONB-CA, respectively). This indicates good cycling stability in practical applications. Additionally, an increase in R_{ct} was observed, which may be associated with the detachment and dissolution of some electroactive substances due to the conductive adhesive.⁷⁰

O-CA, ON-CA, and ONB-CA were utilized as anode materials for further investigation of PIBs.⁷¹ To assess the electrochemical performance of O/N/B-CA, constant current discharge/charge cycling and CV techniques

were utilized. Figure S29 presents the first three consecutive CV curves for the O-CA, ON-CA, and ONB-CA electrodes. The initial CV curve of ON-CA exhibits two prominent reduction peaks at 1.32 and 0.58 V, which correspond to the adsorption of K⁺ and the redox reaction between the active material and K⁺. Additionally, the formation of a solid electrolyte interface (SEI) film on the surface of the carbonaceous matrix is observed. The redox peaks appear at lower potentials and diminish sharply in subsequent cycles. Commencing from the second cycle, the minimal cathodic/anodic peaks and the overlapping CV curves suggest that the primary reactions take place on or in the vicinity of the regions, accompanied by a remarkably reversible de/intercalation process between the carbon layers.⁷² Figure 6A shows the CV curves of ON-CA within the voltage window of 0.01–2.6 V and scan rates of 0.5–5 mV s^{-1} . The irreversible reduction peak observed at 1.85 V on each curve is attributed to an irreversible reaction between K⁺ and the surface functional groups present on ON-CA. Furthermore, the increase in scan rate resulted in a corresponding increase in the oxidation–reduction current density of the samples, signifying exceptional electrochemical performance during the charging and discharging processes. The CV curves of O-CA and ONB-CA are presented in Figure S30. Figure S31 displays the calculated b values of the oxidation peaks for O-CA, ON-CA, and ONB-CA

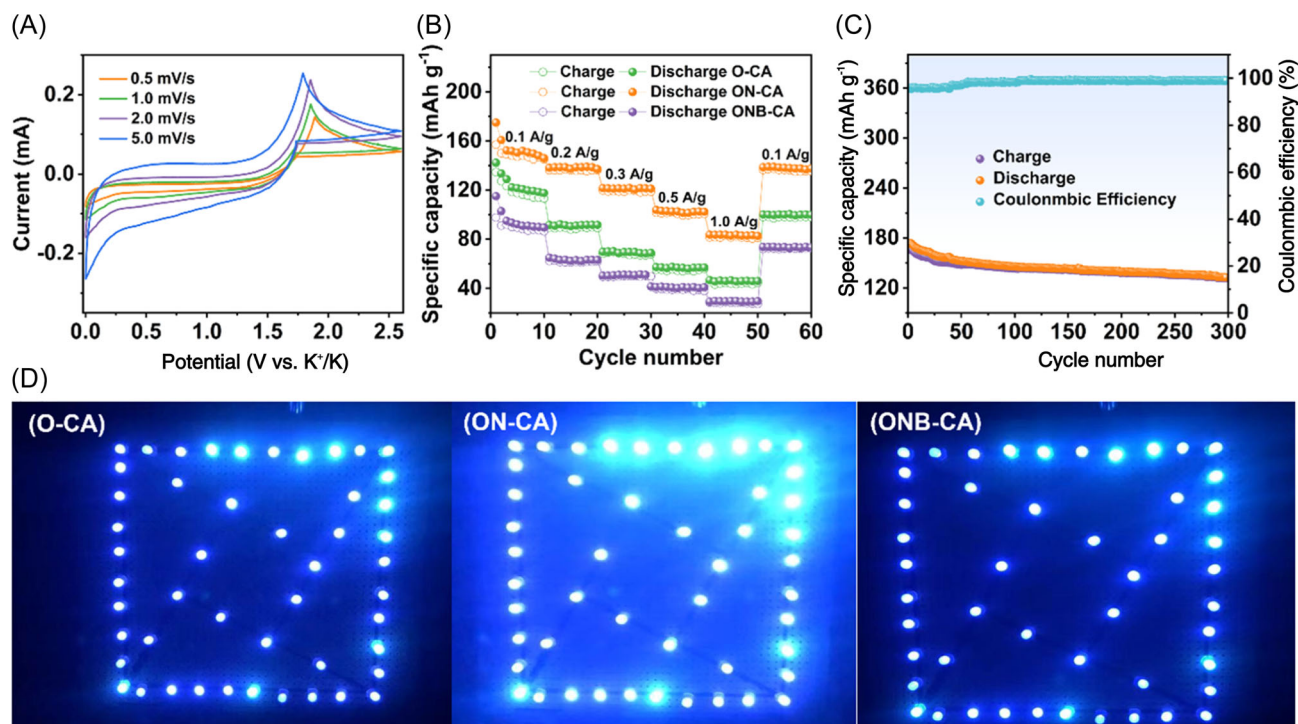


FIGURE 6 (A) CV curves of ON-CA at different scan rates. (B) The rate capabilities of O-CA, ON-CA, and ONB-CA. (C) Electrochemical cycle stability of ON-CA in cycles at 0.1 A g^{-1} . (D) The digital photos of O-CA, ON-CA, and ONB-CA assembled PIBs lighting up blue LEDs.

using Formula (5), which are 7.62×10^{-5} , 9.65×10^{-5} , and 5.79×10^{-5} , respectively. The diffusion coefficients of K^+ for O-CA, ON-CA, and ONB-CA were determined using Formula 1S, resulting in values of 8.6×10^{-16} , 1.10×10^{-15} , and $6.6 \times 10^{-16} \text{ cm}^2 \text{ s}^{-1}$, respectively. These findings demonstrate that the introduction of O and N atoms has a significant impact on the ion diffusion rate during the charging and discharging processes.

To conduct a more in-depth analysis of the constant current charge–discharge behavior of O-CA, ON-CA, and ONB-CA, the charge and discharge curves are presented in Figure S32. At a current density of 0.02 A g^{-1} , the initial discharge and charge capacities of ON-CA are determined to be 376.8 and 174.1 mAh g^{-1} , respectively, while exhibiting an initial Coulombic efficiency (ICE) of 46.2% . In comparison, the ICE values of O-CA and ONB-CA were 23.2% and 38.4% , respectively. The high ICE observed for ON-CA indicates a relatively low specific surface area, which can be attributed to the porous nature, defects, and functional groups. These characteristics result in a reduced formation of SEI film on the ON-CA electrode, which in turn leads to a lower likelihood of electrolyte decomposition.

Figure 6B depicts the rate performance of O-CA, ON-CA, and ONB-CA. Over the first 60 cycles, the specific capacities exhibit a decreasing trend with an increase in current

density. However, upon reverting to a current density of 100 mA g^{-1} after the 51st cycle, the capacities of ON-CA, O-CA, and ONB-CA recovered to 75% , 85% , and 64% of their initial capacities, respectively, indicating that ON-CA exhibited superior rate performance. The observed phenomenon can be primarily ascribed to two factors: (i) the 3D surface of the CA containing oxygen and nitrogen functional groups enhances the wettability of the electrode material with the electrolyte, thereby facilitating the intercalation and deintercalation of a greater number of potassium ions. (ii) The introduction of nitrogen and oxygen heteroatoms through doping creates additional charge-active sites on the surface of the material, which can either donate or accept electrons or holes. The long-term cycling performance of ON-CA at a current density of 0.02 A g^{-1} is depicted in Figure 6C. The specific capacity is maintained at 105 mAh g^{-1} after 300 cycles, surpassing the cycling performance of O-CA and ONB-CA (Figure S33). Simultaneously, the impedance results demonstrate that the charge transfer resistance (R_{ct}) of the ON-CA anode is the lowest among all samples, as shown in Figure S34. These findings suggest that the unique 3D porous CA structure of the ON-CA anode promotes surface wettability and enhances conductivity through the synergistic effect of heteroatom doping. Consequently, the ON-CA anode exhibits an ideal electrochemical architecture, providing ample space and pathways for the rapid and

unhindered diffusion and transfer of potassium ions and electrons. This observation is consistent with previous electrochemical tests.^{73,74} Finally, as illustrated in Figure 6D, O-CA, ON-CA, and ONB-CA were capable of successfully lighting up 49 light-emitting diode (LED) bulbs, demonstrating the significant commercial value of heteroatom-doped CA.

To investigate the N/O/B functional mechanisms for K^+ storage enhancement, DFT calculations were carried out. Three structural models, namely O-CA, ON-CA, and ONB-CA, were introduced via first-principles calculations.⁷⁵ The established and optimized geometrical structures of several models are shown in Figure 7A–C together with the associated K atom adsorption energies. Based on the computed results, it was found that the carbon structure of the ON-CA model, which contains both nitrogen and oxygen

dopants, exhibits smaller adsorption energy for K^+ (−1.62 eV) compared to those of O-CA (−2.16 eV) and ONB-CA (−2.55 eV). As a general rule, a more negative adsorption energy value is indicative of a stronger adsorption capability, which consequently results in greater stability of the CA material. However, in the case of ON-CA, the carbon structure exhibits relatively lower adsorption energy toward potassium ions, resulting in higher diffusivity. It is noteworthy that adsorption energy alone does not entirely dictate the material's performance. Other factors, such as electrical conductivity, ion diffusion coefficient, and reaction activity, may significantly impact the material's overall performance. A visualization of the 3D Bader charge density for all three factors can be found in Figure S35.

The present study employs the density of states (DOS) calculations of O-CA, ON-CA, and ONB-CA models to

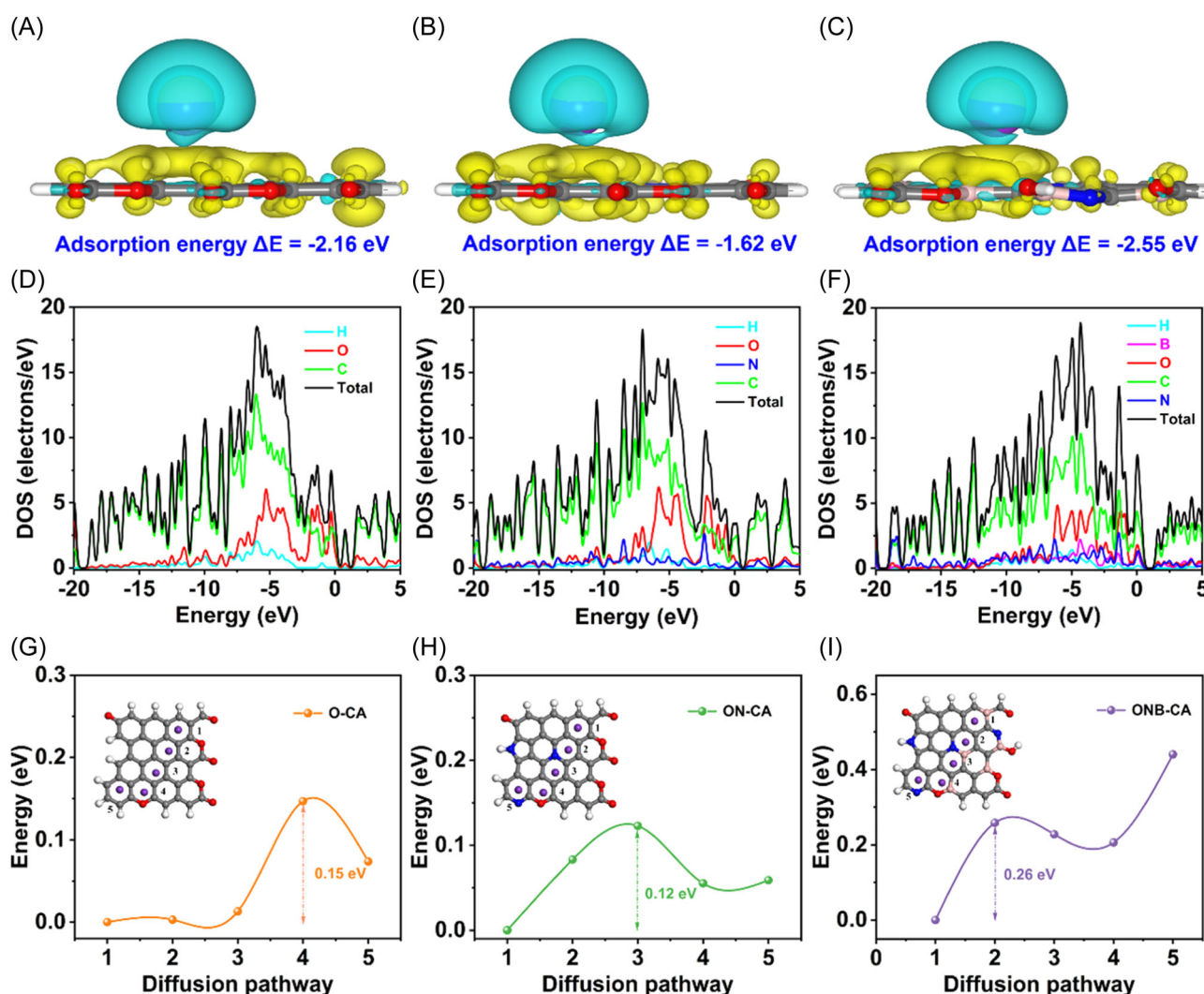


FIGURE 7 First-principles analysis of O-CA, ON-CA, and ONB-CA electrodes: side view of 3D differential charge density map for (A) O-CA, (B) ON-CA, and (C) ONB-CA; DOS for (D) O-CA, (E) ON-CA, and (F) ONB-CA adsorbed with K^+ . Diffusion path of K^+ in (G) O-CA, (H) ON-CA, and (I) ONB-CA. The white, red, gray, and blue spheres, respectively, represent hydrogen, oxygen, carbon, and nitrogen atoms, while the yellow and blue regions correspond to an increase and decrease in electron density, respectively.

explore the adsorption mechanism of potassium ions on the surface of CA.⁷⁶ In Figure 7D–F, it can be observed that the electronic states of ON-CA, and ONB-CA near the Fermi level have increased, indicating that the doping of N and B in CA has enhanced the electron transfer rate, which effectively improves the adsorption and diffusion of K⁺ on the material surface. The inherent DOS of CA in the figures is mainly divided into three parts: the section from −20 to −10 eV corresponds to the sp² hybridized CA with δ bond formation, the section from −10 to 0 eV corresponds to the π orbital of CA, and the section from 0 to −5 eV corresponds to the anti- π orbital of CA. However, ON-CA exhibits the highest total density of states between −20 and −10 eV, indicating the most electronic contributions of the p and s orbitals of the K atom in ON-CA. This is mainly due to the strong electronegativity of nitrogen and oxygen atoms and hydrophilic functional groups of hydroxyl and amino formed by them.⁷⁷

To further investigate the migration behavior of potassium ions on the surfaces of O-CA, ON-CA, and ONB-CA, possible diffusion pathways (as shown in Figure 7G–I) were designed, and the corresponding atomic migration energy curves were calculated. When potassium ions diffuse along the optimal pathway in O-CA, ON-CA, and ONB-CA, the highest diffusion energy barriers and migration activation energies are 0.15, 0.12, and 0.26 eV, respectively. It is evident that the migration activation energy required for potassium ions to migrate on the ON-CA route is smallest, indicating that the ON-CA surface has more functional groups that reduce the surface energy, making it easier for potassium ions to diffuse on the surface.⁷⁸ The optimal pathway of potassium ions in ON-CA is along the carbon ring with N or O doping, indicating that the N and O doping alters the electronic structure of the CA surface, resulting in changes in the band structure.

4 | CONCLUSION

In summary, we present a novel approach to prepare CA using sustainable biomaterial polysaccharides as precursors. The synergistic effect of N and O functionalized CA can enhance the adsorption capacity for K⁺, increase electronic and ionic conductivity, and lower the diffusion barrier, thereby contributing to the outstanding K⁺ storage performance of ON-CA as a cathode material. This work demonstrates that N and O codoped CA is a promising candidate for large-scale K⁺ energy storage applications, and expands the application prospects of low-cost and abundant biomass-based carbon materials in energy storage. Notably, the facile assembly method renders this material a more efficient and durable cathode material in supercapacitors and batteries.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there are no conflict of interests.

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REFERENCES

- Chmiola J, Largeot C, Taberna PL, Simon P, Gogotsi Y. Monolithic carbide-derived carbon films for micro-supercapacitors. *Science*. 2010;328(5977):480–483.
- Kang B, Ceder G. Battery materials for ultrafast charging and discharging. *Nature*. 2009;458(7235):190–193.
- Pachfule P, Shinde D, Majumder M, Xu Q. Fabrication of carbon nanorods and graphene nanoribbons from a metal-organic framework. *Nat Chem*. 2016;8(7):718–724.
- Peng X, Liu H, Yin Q, et al. A zwitterionic gel electrolyte for efficient solid-state supercapacitors. *Nat Commun*. 2016;7(1):11782.
- Zhu Y, Murali S, Stoller MD, et al. Carbon-based supercapacitors produced by activation of graphene. *Science*. 2011;332(6037):1537–1541.
- Borchardt L, Leistenschneider D, Haase J, Dvoyashkin M. Revising the concept of pore hierarchy for ionic transport in carbon materials for supercapacitors. *Adv Energy Mater*. 2018;8(24):1800892.
- Chen Q, Jin J, Song M, et al. High-energy aqueous ammonium-ion hybrid supercapacitors. *Adv Mater*. 2022;34(8):2107992.
- Chen Y, Shi X, Lu B, Zhou J. Concave engineering of hollow carbon spheres toward advanced anode material for sodium/potassium ion batteries. *Adv Energy Mater*. 2022;12(46):2202851.
- Ding M, Li S, Guo L, et al. Metal ion induced assembly of MXene aerogels via biomimetic microtextures for electromagnetic interference shielding, capacitive deionization, and microsupercapacitors. *Adv Energy Mater*. 2021;11(35):2101494.
- Eftekhari A. Supercapacitors utilising ionic liquids. *Energy Storage Mater*. 2017;9:47–69.
- Fan L, Lin K, Wang J, Ma R, Lu B. A nonaqueous potassium-based battery-supercapacitor hybrid device. *Adv Mater*. 2018;30(20):1800804.
- Feng W, Feng N, Liu W, et al. Liquid state templates for constructing B, N, Co doping porous carbons with a boosting of potassium ion storage performance. *Adv Energy Mater*. 2020;11(4):2003215.
- Ganfoud N, Sene A, Haelele M, et al. Effect of the carbon microporous structure on the capacitance of aqueous supercapacitors. *Energy Storage Mater*. 2019;21:190–195.
- Kang MS, Heo I, Cho KG, et al. Coarsening-induced hierarchically interconnected porous carbon polyhedrons for

- stretchable ionogel-based supercapacitors. *Energy Storage Mater.* 2022;45:380-388.
15. Kong S, Xiang X, Jin B, et al. B, O and N codoped biomass-derived hierarchical porous carbon for high-performance electrochemical energy storage. *Nanomaterials.* 2022;12(10):1720.
 16. Li D, Ren X, Ai Q, et al. Facile fabrication of nitrogen-doped porous carbon as superior anode material for potassium-ion batteries. *Adv Energy Mater.* 2018;8(34):1802386.
 17. Lin X, Yin S, Jiang J, Li X. B/N/O/Zn doped porous carbon materials for supercapacitor with high performance. *J Electroanal Chem.* 2022;918:116498.
 18. Liu Y, Dai H, Wu L, et al. A large scalable and low cost sulfur/nitrogen dual doped hard carbon as the negative electrode material for high performance potassium ion batteries. *Adv Energy Mater.* 2019;9(34):1901379.
 19. Liu Y, Lu YX, Xu YS, et al. Pitch-derived soft carbon as stable anode material for potassium ion batteries. *Adv Mater.* 2020;32(17):2000505.
 20. Ruan J, Mo F, Chen Z, et al. Rational construction of nitrogen doped hierarchical dual carbon for advanced potassium ion hybrid capacitors. *Adv Energy Mater.* 2020;10(15):1904045.
 21. Wu Y, Zhao H, Wu Z, et al. Rational design of carbon materials as anodes for potassium-ion batteries. *Energy Storage Mater.* 2021;34:483-507.
 22. Yuan W, Liu J, Yi W, Liang L, Zhu Y, Chen X. Boron and nitrogen co-doped double-layered mesopore-rich hollow carbon microspheres as high-performance electrodes for supercapacitors. *J Colloid Interface Sci.* 2020;573:232-240.
 23. Zhang C. Super pseudocapacitors. *Nat Energy.* 2018;3(12):1019.
 24. Yang X, Gong L, Wang K, et al. Ionothermal synthesis of fully conjugated covalent organic frameworks for high-capacity and ultrastable potassium-ion batteries. *Adv Mater.* 2022;34(50):2207245.
 25. Zhang X, Kar M, Mendes TC, Wu Y, MacFarlane DR. Supported ionic liquid gel membrane electrolytes for flexible supercapacitors. *Adv Energy Mater.* 2018;8(15):1702702.
 26. Zhu Q, Zhao D, Cheng M, et al. A new view of supercapacitors: integrated supercapacitors. *Adv Energy Mater.* 2019;9(36):1901081.
 27. Cui RC, Xu B, Dong HJ, Yang CC, Jiang Q. N/O dual-doped environment-friendly hard carbon as advanced anode for potassium-ion batteries. *Adv Sci.* 2020;7(5):1902547.
 28. Zhu Y, Wang Y, Wang Y, Xu T, Chang P. Research progress on carbon materials as negative electrodes in sodium and potassium-ion batteries. *Carbon Energy.* 2022;4(6):1182-1213.
 29. Khossossi N, Luo W, Haman Z, et al. Revealing the superlative electrochemical properties of o-B₂N₂ monolayer in lithium/sodium-ion batteries. *Nano Energy.* 2022;96:107066.
 30. Chen H, Liu T, Mou J, et al. Free-standing N-self-doped carbon nanofiber aerogels for high-performance all-solid-state supercapacitors. *Nano Energy.* 2019;63:103836.
 31. Cho KG, Kim HS, Jang SS, et al. Optimizing electrochemically active surfaces of carbonaceous electrodes for ionogel based supercapacitors. *Adv Funct Mater.* 2020;30(30):2002053.
 32. Kim CHJ, Zhao D, Lee G, Liu J. Strong, machinable carbon aerogels for high performance supercapacitors. *Adv Funct Mater.* 2016;26(27):4976-4983.
 33. Liu X, Taiwo OO, Yin C, et al. Aligned ionogel electrolytes for high-temperature supercapacitors. *Adv Sci.* 2019;6(5):1801337.
 34. Wang P, Zhang G, Li M-Y, et al. Porous carbon for high-energy density symmetrical supercapacitor and lithium-ion hybrid electrochemical capacitors. *Chem Eng J.* 2019;375:122020.
 35. Wei X, Wan S, Gao S. Self-assembly-template engineering nitrogen-doped carbon aerogels for high-rate supercapacitors. *Nano Energy.* 2016;28:206-215.
 36. Biener J, Stadermann M, Suss M, et al. Advanced carbon aerogels for energy applications. *Energy Environ Sci.* 2011;4(3):656-667.
 37. Guo F, Jiang Y, Xu Z, et al. Highly stretchable carbon aerogels. *Nat Commun.* 2018;9(1):881.
 38. Grimme S, Antony J, Ehrlich S, Krieg H. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu. *J Chem Phys.* 2010;132(15):154104.
 39. Kresse G, Joubert D. From ultrasoft pseudopotentials to the projector augmented-wave method. *Phys Rev B.* 1999;59(3):1758-1775.
 40. Kresse G, Furthmüller J. Efficient iterative schemes for ab initio total energy calculations using a plane-wave basis set. *Phys Rev B.* 1996;54(16):11169-11186.
 41. Li C, Kong D, Wang B, et al. Conjugated microporous polymer derived N, O and S co-doped sheet-like carbon materials as anode materials for high-performance lithium-ion batteries. *J Taiwan Inst Chem Eng.* 2022;134:104293.
 42. Jürgens B, Irran E, Senker J, Kroll P, Müller H, Schnick W. Melem (2,5,8-triamino-tri-s-triazine), an important intermediate during condensation of melamine rings to graphitic carbon nitride: synthesis, structure determination by X-ray powder diffractometry, solid-state NMR, and theoretical studies. *J Am Chem Soc.* 2003;125(34):10288-10300.
 43. Deng W, Hu M, Xu S, et al. Pyrolysis of sludge briquettes for the preparation of cylindrical-shaped biochar and comparison between CO₂ and steam activation. *Fuel.* 2023;338:127317.
 44. Worsley MA, Pauzauskie PJ, Olson TY, Biener J, Satcher JH, Baumann TF. Synthesis of graphene aerogel with high electrical conductivity. *J Am Chem Soc.* 2010;132(40):14067-14069.
 45. Lai H, Zhuo H, Hu Y, et al. Anisotropic carbon aerogel from cellulose nanofibers featuring highly effective compression stress transfer and pressure sensing. *ACS Sustainable Chem Eng.* 2021;9(29):9761-9769.
 46. Liu Q, Shen M, Duan C, Zhang L, Zhu J, Ni Y. 3D hierarchical porous carbon aerogel electrocatalysts based on cellulose/aramid nanofibers and application in high-performance Zn–air batteries. *ACS Appl Energy Mater.* 2022;5(12):15146-15154.
 47. Zhu J, Sakaushi K, Clavel G, Shalom M, Antonietti M, Feller TP. A general salt-templating method to fabricate vertically aligned graphitic carbon nanosheets and their metal carbide hybrids for superior lithium ion batteries and water splitting. *J Am Chem Soc.* 2015;137(16):5480-5485.
 48. Yan Y, Zhang P, Qu Z, et al. Carbon/sulfur aerogel with adequate mesoporous channels as robust polysulfide confinement matrix for highly stable lithium–sulfur battery. *Nano Lett.* 2020;20(10):7662-7669.
 49. Wu X, Jiang L, Long C, Fan Z. From flour to honeycomb-like carbon foam: carbon makes room for high energy density supercapacitors. *Nano Energy.* 2015;13:527-536.

50. Yu ZL, Xin S, You Y, et al. Ion-catalyzed synthesis of microporous hard carbon embedded with expanded nanographite for enhanced lithium/sodium storage. *J Am Chem Soc*. 2016;138(45):14915-14922.
51. Li SC, Hu BC, Ding YW, et al. Wood-derived ultrathin. *Angew Chem Int Ed*. 2018;57(24):7085-7090.
52. Zhao Y-P, Xu R-X, Cao J-P, Zhang X-Y, Zhu J-S, Wei X-Y. N/O co-doped interlinked porous carbon nanoflakes derived from soybean stalk for high-performance supercapacitors. *J Electroanal Chem*. 2020;871(1):114288.
53. Wu H, Yuan W, Zhao Y, Han D, Yuan X, Cheng L. B, N-dual doped sisal-based multiscale porous carbon for high-rate supercapacitors. *RSC Adv*. 2019;9(3):1476-1486.
54. Yang Z, Fan Q, Lai S, et al. Preparation of N/O-codoped quinoline pitch-based porous carbons for high-quality supercapacitor electrodes. *New J Chem*. 2022;46(11):5266-5277.
55. Han J, Li Q, Wang J, et al. Heteroatoms (O, N)-doped porous carbon derived from bamboo shoots shells for high performance supercapacitors. *J Mater Sci Mater Electron*. 2018;29(24):20991-21001.
56. Zhang J, Chen H, Bai J, et al. N-doped hierarchically porous carbon derived from grape marcs for high-performance supercapacitors. *J Alloys Compd*. 2021;854(1):157207.
57. Dai P, Xue Y, Zhang S, et al. Paper-derived flexible 3D interconnected carbon microfiber networks with controllable pore sizes for supercapacitors. *ACS Appl Mater Interfaces*. 2018;10(43):37046-37056.
58. Dai X, Zheng L, Tang B, Peng J, Chen H. Notoginseng-derived B/N co-doped porous carbon with high N-doped content and good electrochemical performance. *Ionics*. 2021;27(4):1439-1449.
59. Guo D, Ding B, Hu X, Wang Y, Han F, Wu X. Synthesis of boron and nitrogen codoped porous carbon foam for high performance supercapacitors. *ACS Sustainable Chem Eng*. 2018;6(9):11441-11449.
60. Qian X, Miao L, Jiang J, et al. Hydrangea-like N/O codoped porous carbons for high-energy supercapacitors. *Chem Eng J*. 2020;388:124208.
61. Li P, Wang W, Su F, Wang X, Zhang X, Zheng X. N-doped interconnected porous graphene as advanced electrode material for supercapacitors. *J Alloys Compd*. 2022;893:162218.
62. Lin X, Yin S, Zhang W, Li X. N/P/O doped porous carbon materials for supercapacitor with high performance. *Diamond Relat Mater*. 2022;125:109025.
63. Fu R, Yu C, Li S, et al. A closed-loop and scalable process for the production of biomass-derived superhydrophilic carbon for supercapacitors. *Green Chem*. 2021;23(9):3400-3409.
64. Miao L, Duan H, Zhu D, et al. Boron "gluing" nitrogen heteroatoms in a prepolymerized ionic liquid-based carbon scaffold for durable supercapacitive activity. *J Mater Chem A*. 2021;9(5):2714-2724.
65. Yao Y, Huang G, Liu Y, et al. Facile synthesis of B/N co-doped porous carbon nanosheets with high heteroatom content and electrical conductivity for excellent-performance supercapacitors. *Appl Surf Sci*. 2022;580:152236.
66. Fu X, Guo W, Chen L, et al. Coal-derived N,O co-doped mesoporous carbon as electrode material for high performance aqueous electric-double layer capacitors and zinc-ion hybrid supercapacitors. *Electrochim Acta*. 2022;439:141576.
67. Wang J, Xu Z, Eloi JC, Titirici MM, Eichhorn SJ. Ice templated, sustainable carbon aerogels with hierarchically tailored channels for sodium and potassium ion batteries. *Adv Funct Mater*. 2022;32(16):2110862.
68. Thakur AK, Kurtyka K, Majumder M, et al. Recent advances in boron and nitrogen doped carbon based materials and their various applications. *Adv Mater Interfaces*. 2022;9(11):2101964.
69. Yu ZL, Li GC, Fechner N, et al. Polymerization under hypersaline conditions: a robust route to phenolic polymer-derived carbon aerogels. *Angew Chem Int Ed*. 2016;55(47):14623-14627.
70. Wei S, Wan C, Zhang L, et al. N-doped and oxygen vacancy-rich NiCo₂O₄ nanograss for supercapacitor electrode. *Chem Eng J*. 2022;429:132242.
71. Yu ZL, Qin B, Ma ZY, et al. Superelastic hard carbon nanofiber aerogels. *Adv Mater*. 2019;31(23):1900651.
72. Deng H, Wang L, Li S, et al. Radial pores in nitrogen/oxygen dual doped carbon nanospheres anode boost high power and ultrastable potassium ion batteries. *Adv Funct Mater*. 2021;31(51):2107246.
73. Ou H, Huang J, Zhou Y, et al. Surface-dominated ultra-stable sodium and potassium storage enabled by N/P/O tri-doped porous carbon. *Chem Eng J*. 2022;450:138444.
74. Wu L, Fu H, Li S, et al. Phase-engineered cathode for super-stable potassium storage. *Nat Commun*. 2023;14(1):644.
75. Nie Z, Zhang L, Ding X, et al. Catalytic kinetics regulation for enhanced electrochemical nitrogen oxidation by Ru-nanoclusters-coupled Mn₃O₄ catalysts decorated with atomically dispersed Ru atoms. *Adv Mater*. 2022;34(14):2108180.
76. Guo M, Fang L, Zhang L, et al. Pulsed electrocatalysis enabling high overall nitrogen fixation performance for atomically dispersed Fe on TiO₂. *Angew Chem Int Ed*. 2023;62(13):e202217635.
77. Deng X, Tian Y, Zou K, et al. K_xC_y phase induced expanded interlayer in ultra-thin carbon toward full potassium-ion capacitors. *Carbon Energy*. 2022;4(6):1151-1168.
78. Cheng G, Zhang W, Wang W, et al. Sulfur and nitrogen codoped cyanoethyl cellulose-derived carbon with superior gravimetric and volumetric capacity for potassium ion storage. *Carbon Energy*. 2022;4(5):986-1001.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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